# Build the Recommender

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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## Recall and choose your starting point

> Record your starting explanation for Build the Recommender before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your **explanation**.

### Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## Practise with fading help

> Complete the supported Build the Recommender trace and record the first step you can explain without help.

Use W2: Handle no match for Implementing a **recommender**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

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## Individual reasoning check

> Check the proposed Build the Recommender explanation against the supplied facts before discussing it.

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Changed independent case

Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated.

Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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## Your learning target

> Connect the Build the Recommender target to a trace and a justified conclusion.

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

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## Solve the given case

> Solve the supplied Build the Recommender case and show the deciding evidence.

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and **evidence** to display.

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

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## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Build the Recommender adds to Transparent Book Recommender and keep the evidence.

This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

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## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

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## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

**My independently reasoned answer**

Record support used: none, hint or teacher **explanation**.

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## W1 Trace selection

Trace selection. Apply the supplied Build the Recommender facts and keep your reasoning.

Why does A win the worked example even though B is **process**ed first?

**My reasoning and evidence**

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## W2 Handle no match

> Handle no match. Apply the supplied Build the Recommender facts and keep your reasoning.

State recommend([], 10) and the worked items with limit 8.

**My reasoning and evidence**

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## W3 Repair order dependence

> Repair order dependence. Apply the supplied Build the Recommender facts and keep your reasoning.

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

**My reasoning and evidence**

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## W4 Implement independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Implement independently. Apply the supplied Build the Recommender facts and keep your reasoning.

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

> **My reasoning and evidence**

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## W5 Explain the result

> Explain the result. Apply the supplied Build the Recommender facts and keep your reasoning.

Write a one-sentence **explanation** of X winning without calling it objectively best.

> **My reasoning and evidence**

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## Building the data pipeline — W1 Trace a validation

> Use the supplied Build the Recommender task to record your reasoning, evidence and next step.

Explain why a row with fun "4" fails while fun 4 can pass.

**My reasoning and evidence**

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## Building the data pipeline — W2 Check missing fields

> Use the supplied Build the Recommender task to record your reasoning, evidence and next step.

What does valid_item return if learning is absent?

**My reasoning and evidence**

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## Building the data pipeline — W3 Repair the connection

> Use the supplied Build the Recommender task to record your reasoning, evidence and next step.

A program validates rows but still sends the original raw list to recommend. What should change?

**My reasoning and evidence**

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## Building the data pipeline — W4 Build independently

> Use the supplied Build the Recommender task to record your reasoning, evidence and next step.

Combine validation and the Week 24 selector for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and **explanation**.

**My reasoning and evidence**

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## Building the data pipeline — W5 Handle duplicates

Use the supplied Build the Recommender task to record your reasoning, evidence and next step.

Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

**My reasoning and evidence**

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## Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 23 OF 26**

## Pictorial case practice: Build the Recommender

> A fictional book chooser filters a request before comparing preference scores. Explain the labelled step before checking the changed condition.

A fictional book chooser filters a request before comparing preference scores.

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

AI ACADEMY / AI-2/33

# Filtering before scoring

| | Pages | Preference score | Status |
|---|---|---|---|
| A | 90 | 2 | eligible |
| B | 70 | 1 | eligible |
| C | 60 | 3 | subject fails |
| D | 80 | 3 | eligible |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/33

## Build the Recommender

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Filter

A,B,D meet subject and page requirements; C fails subject.

### 2. Compute scores

A=2; B=1; D=3.

### 3. Choose best

Return D with both preference matches recorded.

### 4. Deterministic output

Reordering the input rows must not change D's unique winning score.

**CHANGE ONE THING**

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B does not win after D is excluded even though it has fewer pages.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book **Recommender**

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book chooser filters a request before comparing preference scores.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Reordering the **input** rows must not change D's unique winning score.

1. Choose best

2. Filter

3. **Deterministic output**

4. Compute scores

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the **stage** that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported **explanation**: **Deterministic output** Reordering the **input** rows must not change D's unique winning score. Eligibility and ranking answer different questions.

C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner.

Save rejected reasons for an **explanation**, not just the final ID.

### Practice 2: explain the deciding evidence

#### Changed case · predict first

Changed-case practice: Change D's pages to 91, leaving every other field unchanged. Which conclusion is supported by the supplied facts?

1. D becomes ineligible, so A wins with score 2.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

#### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported **explanation**: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

### Repair a missing building block

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

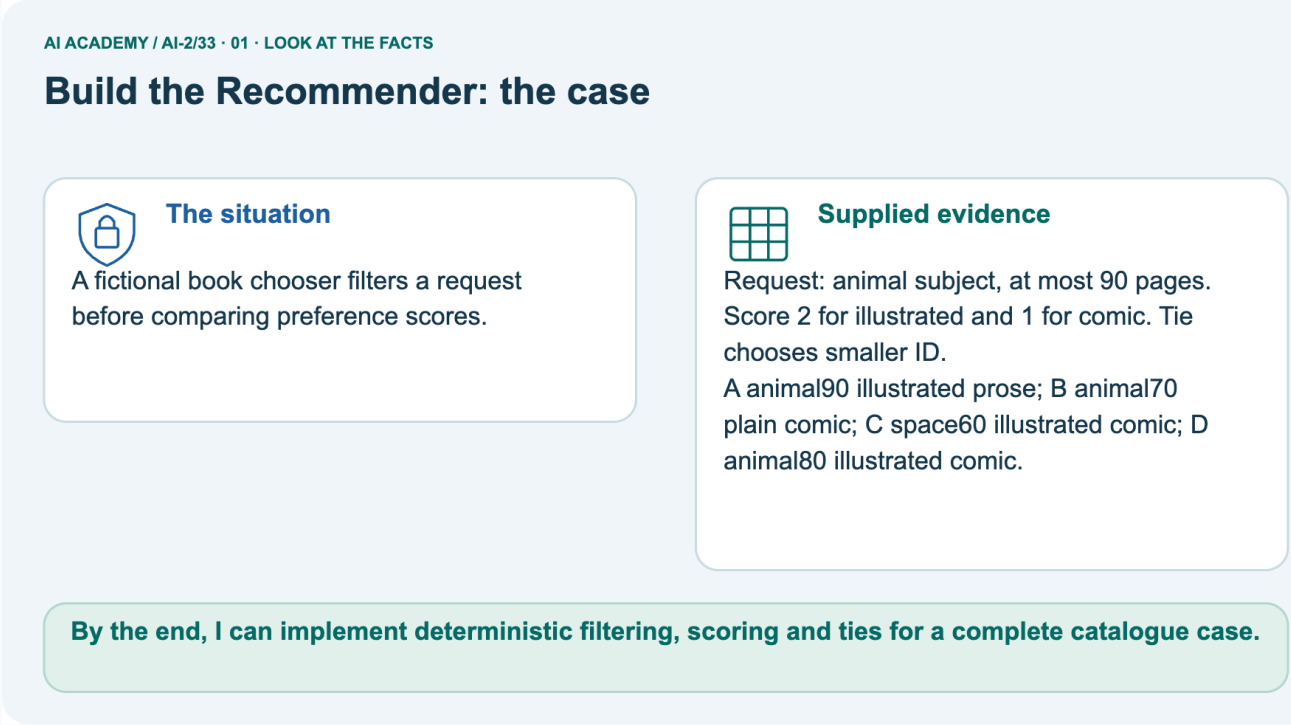
**What I repaired, the evidence I kept and the next test**

WORKBOOK · 26 OF 26

# Visual concept journey

> Explain each Build the Recommender picture from its labels, then record a relevant limit.

## 01. Build the Recommender: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

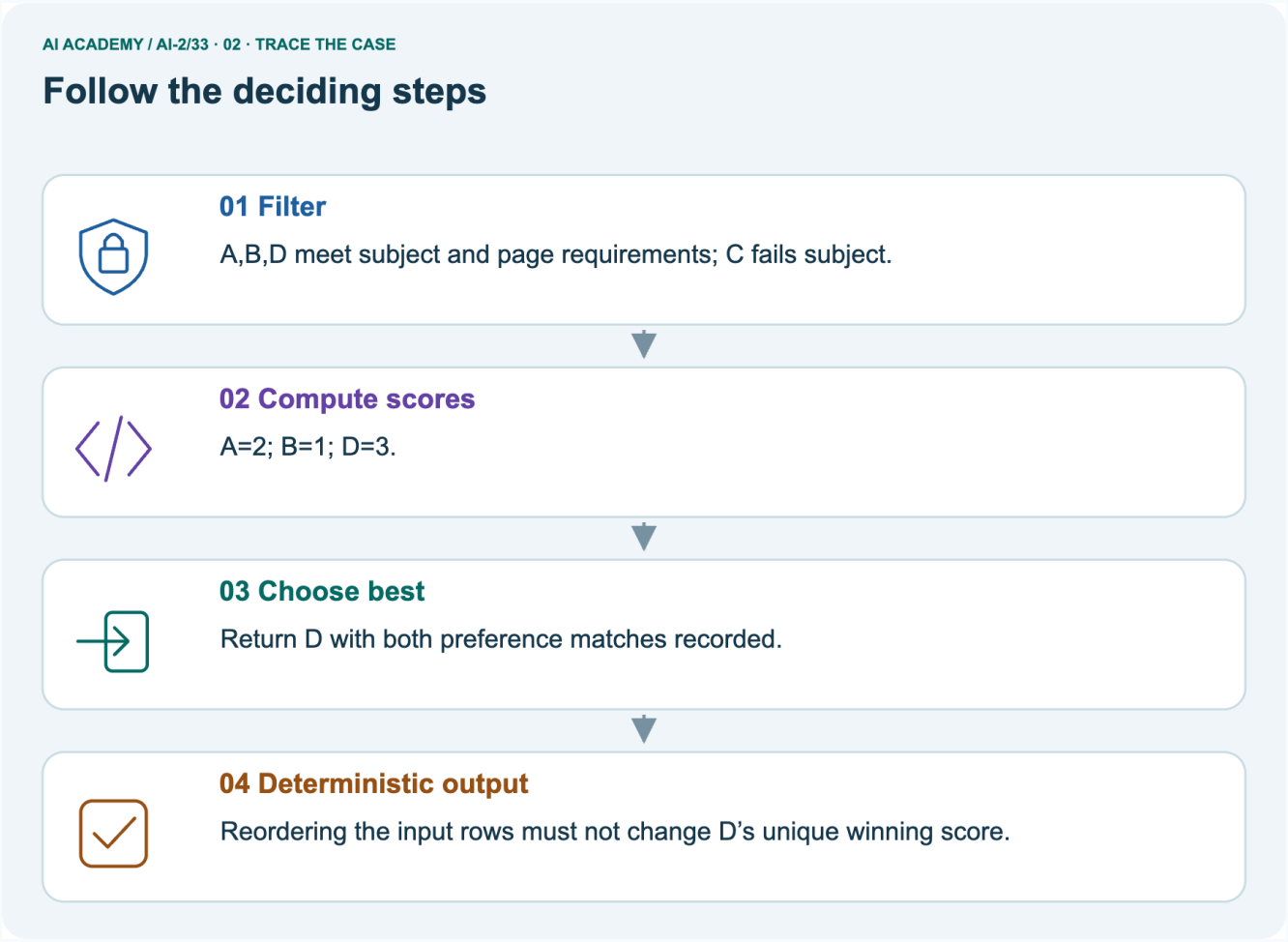
A fictional book chooser filters a request before comparing preference scores. Supplied facts: Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



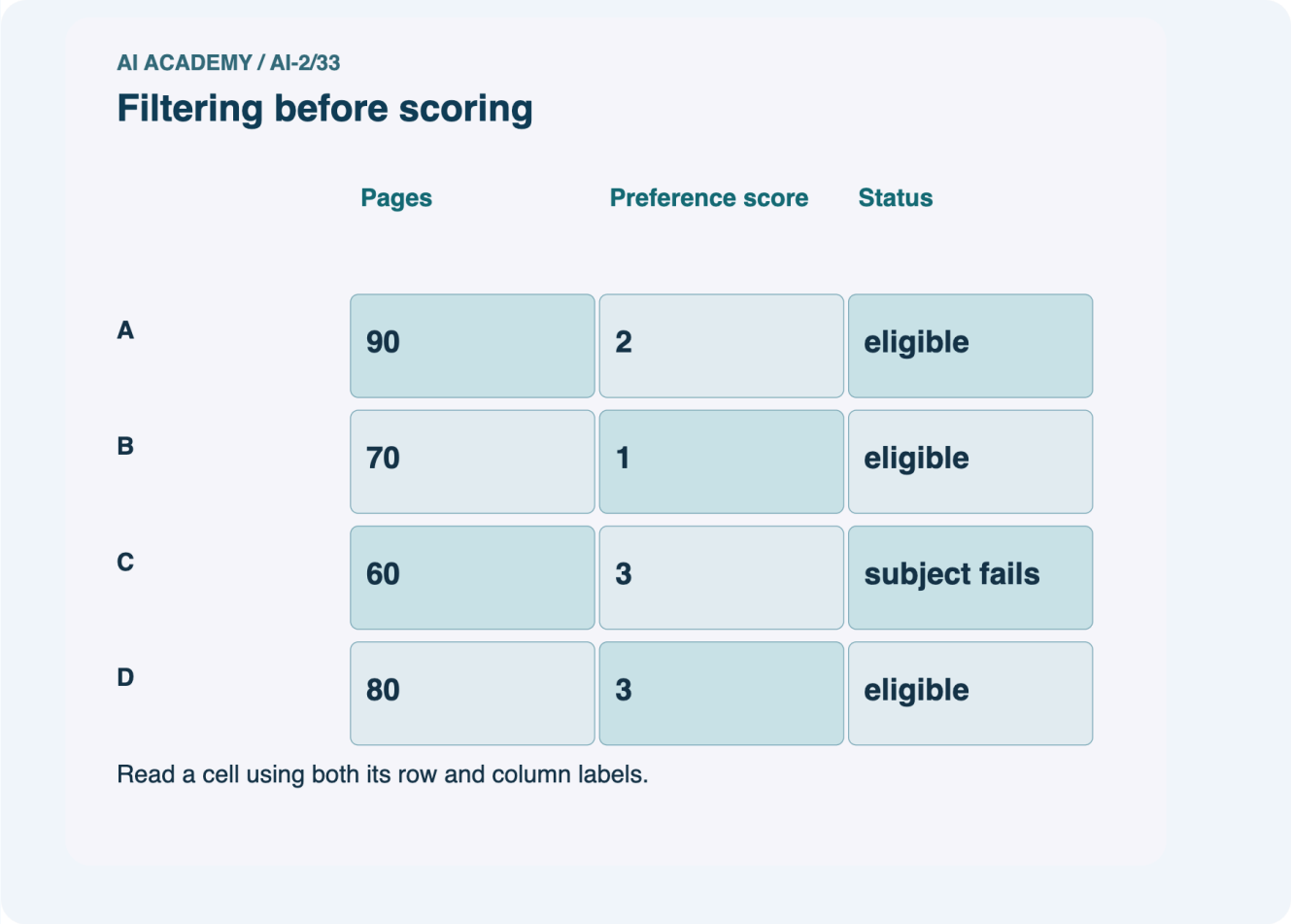

Follow the labelled picture with the matching explanation. Open the vector diagram

Filter: A,B,D meet subject and page requirements; C fails subject. → Compute scores: A=2; B=1; D=3. → Choose best: Return D with both preference matches recorded. → **Deterministic output**: Reordering the **input** rows must not change D's unique winning score.

> Point to each numbered **stage** and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

## 03. Filtering before scoring




Follow the labelled picture with the matching explanation. Open the vector diagram

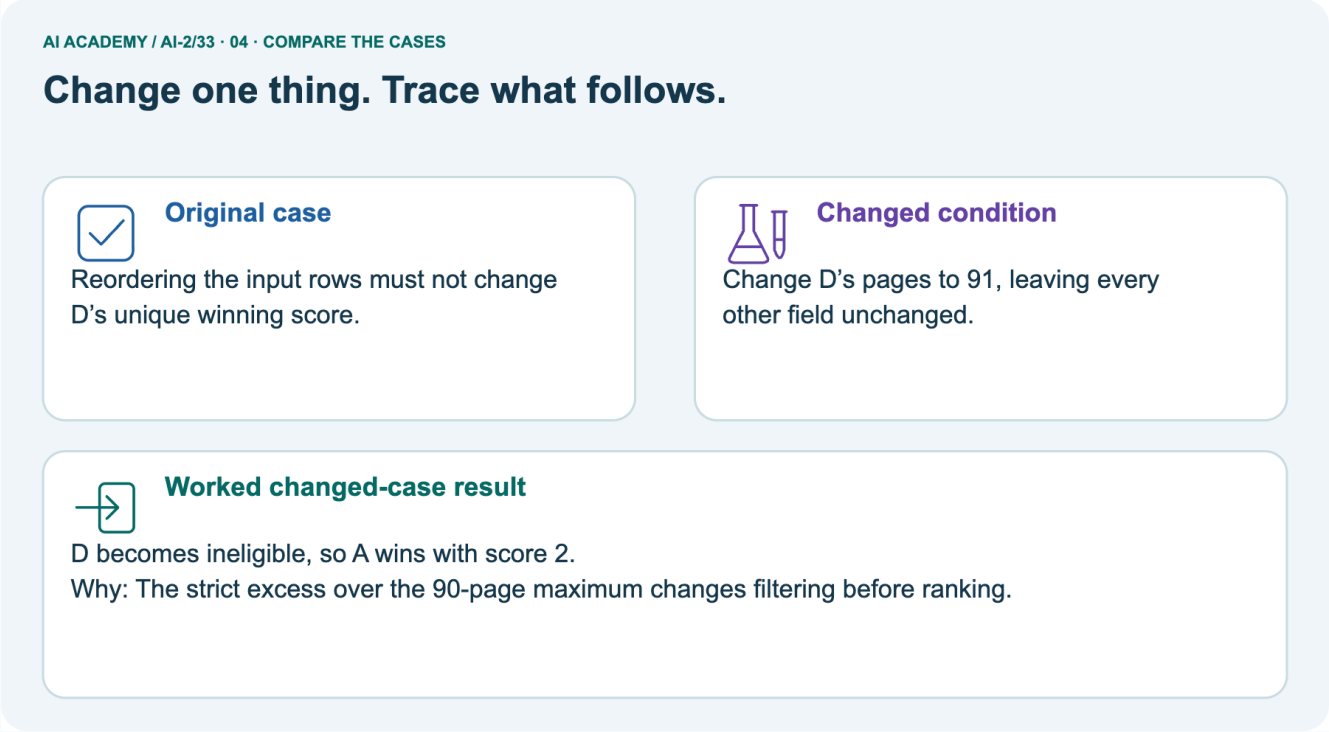
Filtering before scoring. A: Pages = 90, Preference score = 2, Status = eligible. B: Pages = 70, Preference score = 1, Status = eligible. C: Pages = 60, Preference score = 3, Status = subject fails.

D: Pages = 80, Preference score = 3, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

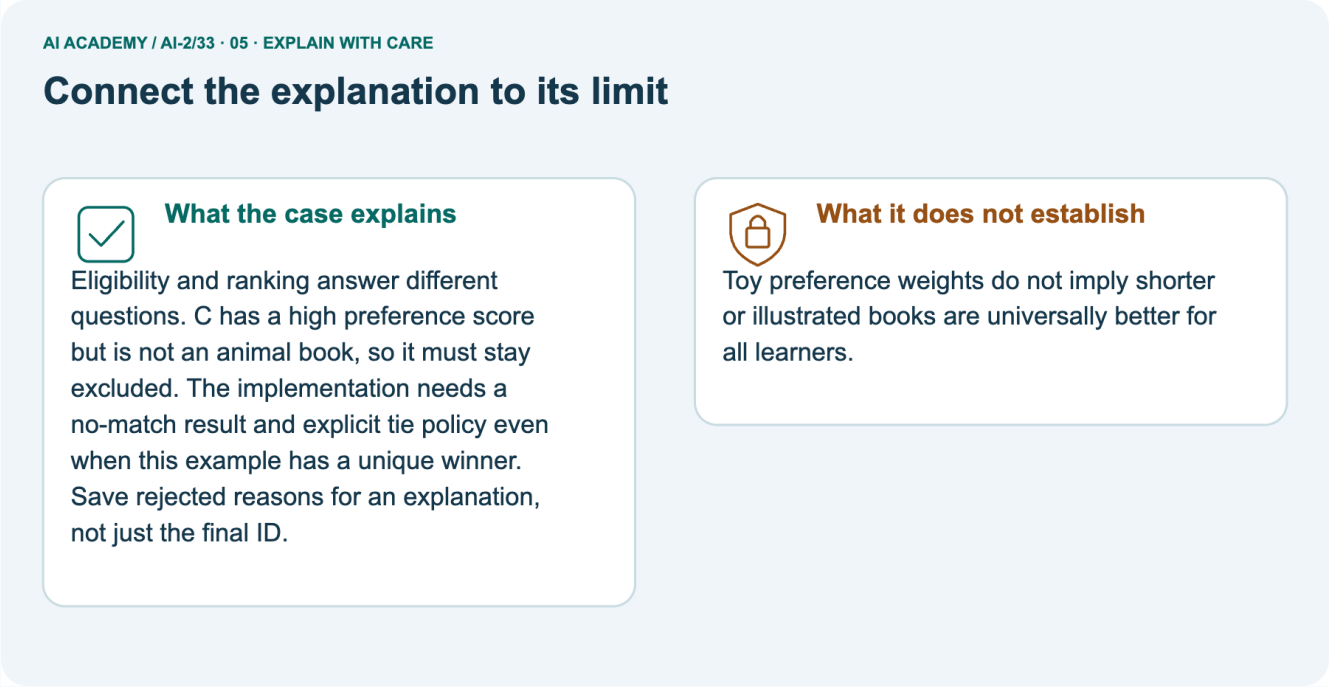
Original case: Reordering the **input** rows must not change D's unique winning score. Changed condition: Change D's pages to 91, leaving every other field unchanged. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

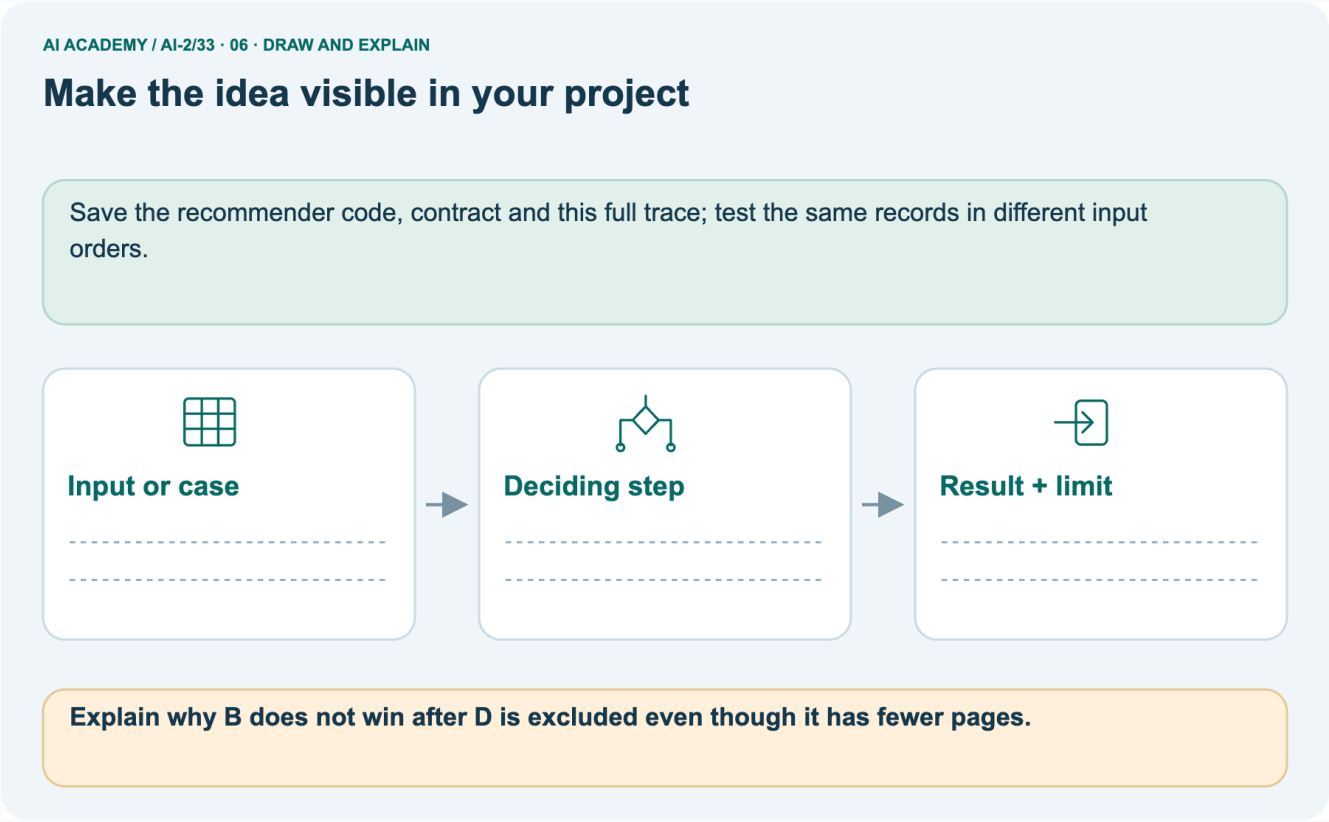
The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

Limit: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

> Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B does not win after D is excluded even though it has fewer pages.

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.